# Inference often combines conditioning and summarizing

**Observed** $y$ $\rightarrow$ **Condition** $p(z \mid y) = \mathcal{N}\left(z \mid \mu_{z|y}, \Sigma_{z|y}\right)$

Murphy, Book 2 §2.3.2.2–2.3.2.3

- Conditioning preserves the whole posterior distribution and its uncertainty

# Inference often combines conditioning and summarizing

| **Observed** | | **Condition** | | **Summarize** |
|---|---|---|---|---|
| $y$ | $\rightarrow$ | $p(z \mid y) = \mathcal{N}\left(z \mid \mu_{z\mid y}, \Sigma_{z\mid y}\right)$ | $\rightarrow$ | $\hat{z} = E[z \mid y] = \mu_{z\mid y}$ |

Murphy, Book 2 §2.3.2.2–2.3.2.3

$$E[z \mid y] = \int z p(z \mid y)\, \mathrm{d}z$$

- Conditioning preserves the whole posterior distribution and its uncertainty
- Summarizing extracts a quantity from that posterior; for a Gaussian the mean is $\mu_{z\mid y}$

# Choose a reconstruction by its conditional squared-error risk

**Observed $y$; choose $a \in \mathbb{R}^d$ to reconstruct unknown $z \in \mathbb{R}^d$**

**Squared-error criterion**

$$L(a, z) = \|z - a\|^2, \quad R(a|y) = E[L(a, z)|y]$$

- Choose $a$ to minimize conditional expected squared error $R(a|y)$.

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**Center at the conditional mean**

$$m = E[z|y], \quad r = z - m, \quad z - a = r + (m - a)$$

- Choose $a$ to minimize conditional expected squared error $R(a|y)$.
- Given $y$, offset $m - a$ is fixed; uncertainty remains in $r$.

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$$\|z - a\|^2 = \|r\|^2 + 2r^T(m - a) + \|m - a\|^2$$

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# Centering the error makes the posterior mean optimal

**Average the centered expansion** Previous slide: pointwise squared-error expansion

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**Take $E[\cdot\,|y]$; $m-a$ is fixed**

$$E[r|y] = E[z|y] - m = 0$$

$$E\left[r^T(m-a)|y\right] = E[r|y]^T(m-a) = 0$$

$$V = \mathrm{Cov}[z|y]$$

$$E\left[\|r\|^2|y\right] = \sum_{j=1}^{d} E\left[r_j^2|y\right] = \mathrm{tr}\, V$$

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- Risk is residual uncertainty plus squared distance from the posterior mean.

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**Conditional squared-error risk**

$$R(a|y) = \mathrm{tr}\, V + \|m-a\|^2$$
$$\|m-a\|^2 \geq 0 \quad \text{equality iff} \quad a = m$$
$$a^*(y) = m = E[z|y]$$
$$\min_a R(a|y) = \mathrm{tr}\, V$$

**Earlier Gaussian**

$$a^*(y) = \mu_{z|y}$$
$$\min_a R(a|y) = \mathrm{tr}\, \Sigma_{z|y}$$

- Risk is residual uncertainty plus squared distance from the posterior mean.
- The posterior mean minimizes model expected squared error; covariance measures what remains.

# Sensor fusion combines measurements according to their uncertainty

**Two noisy measurements of one state**

$$z \sim \mathcal{N}(\mu_z, \Sigma_z)\; x \mid z \sim \mathcal{N}(z, \Sigma_x), \quad y \mid z \sim \mathcal{N}\left(z, \Sigma_y\right)$$

**Same two operations**

$$p(z \mid x, y) \quad \rightarrow \quad E[z \mid x, y]$$

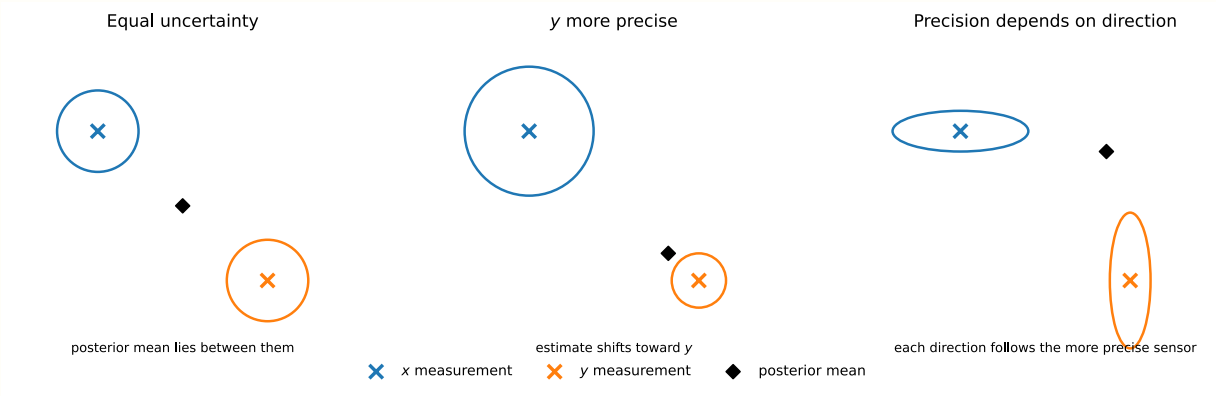
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Course schematic after Murphy, Book 2 §2.3.2.3, Fig. 2.10; `sensor_fusion_figure.ipynb`

- Equal uncertainty puts the posterior mean between the measurements

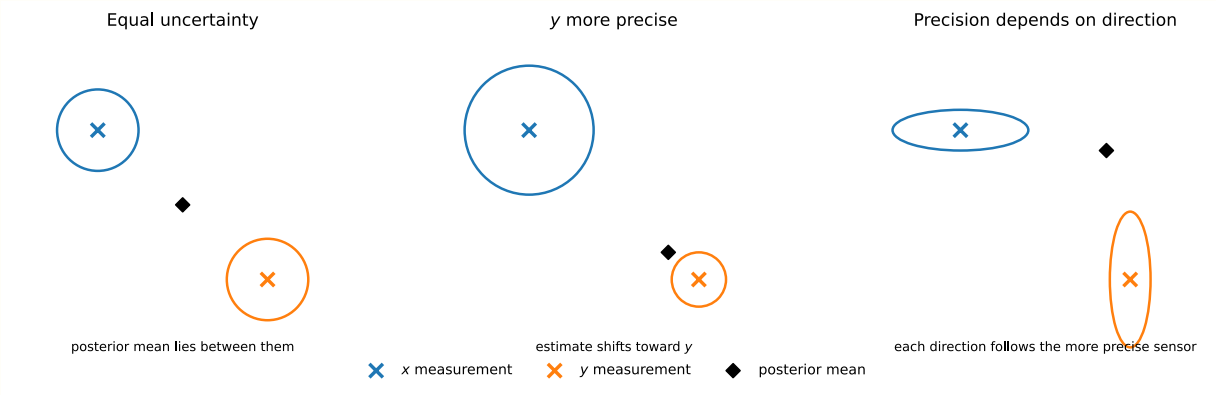
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- A more precise sensor pulls the posterior mean closer to its measurement

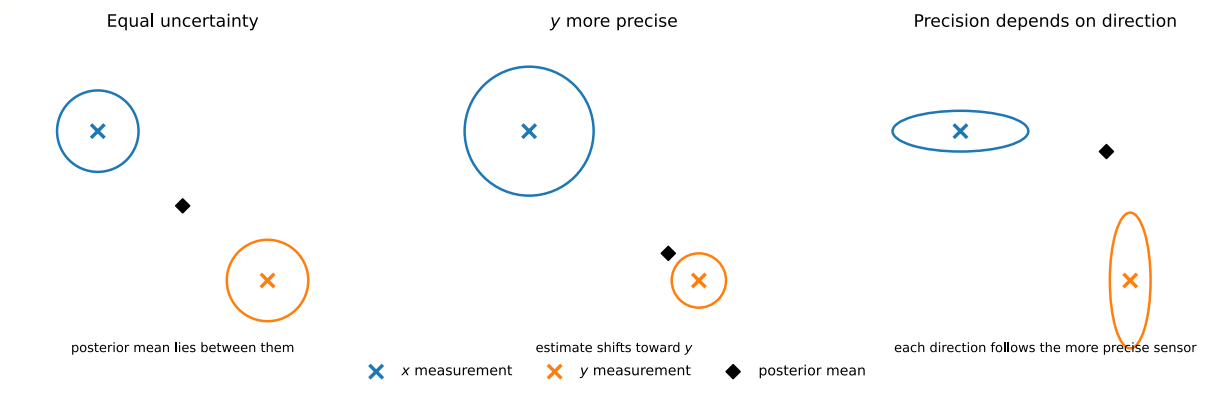
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- With anisotropic uncertainty, the weighting can differ by direction